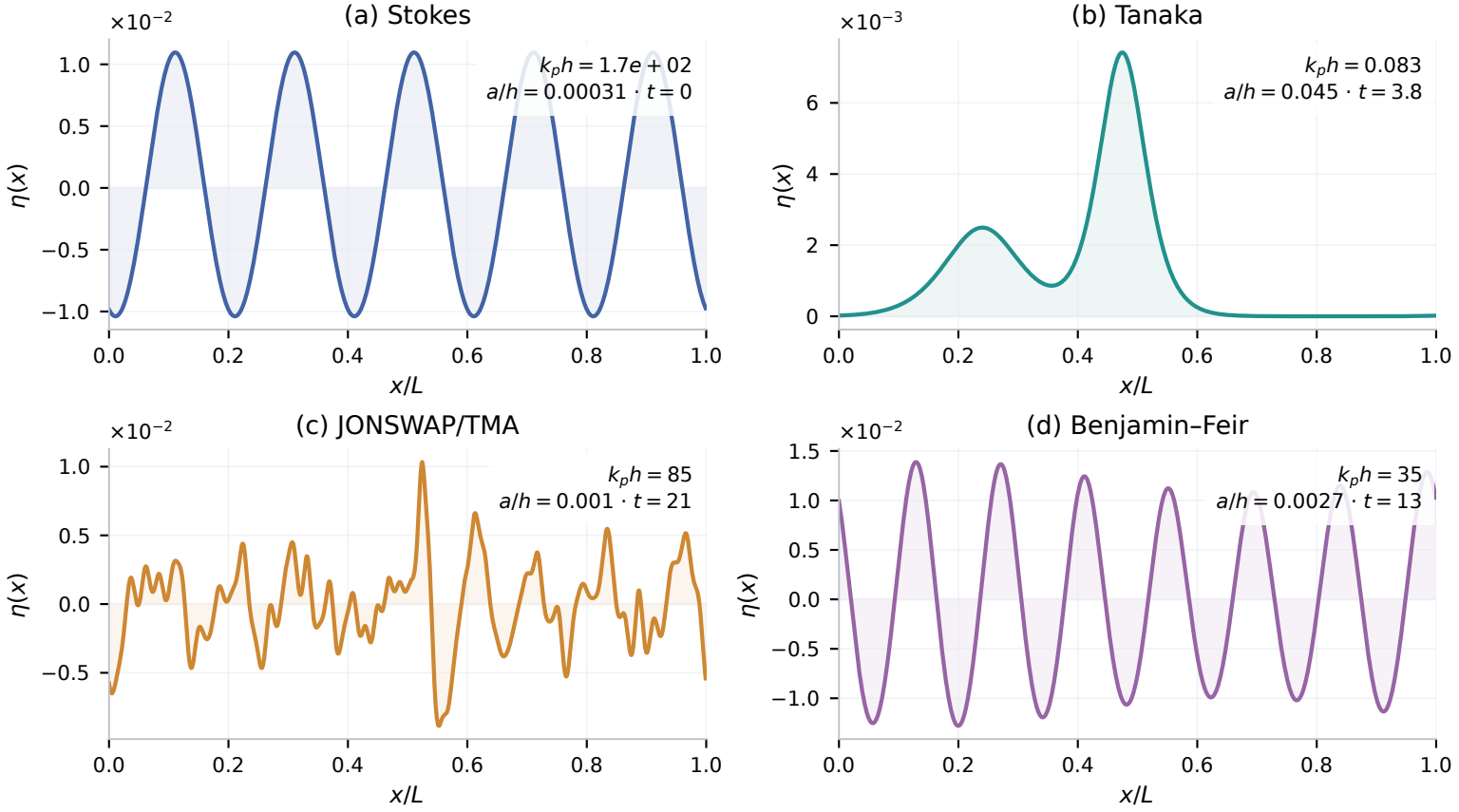
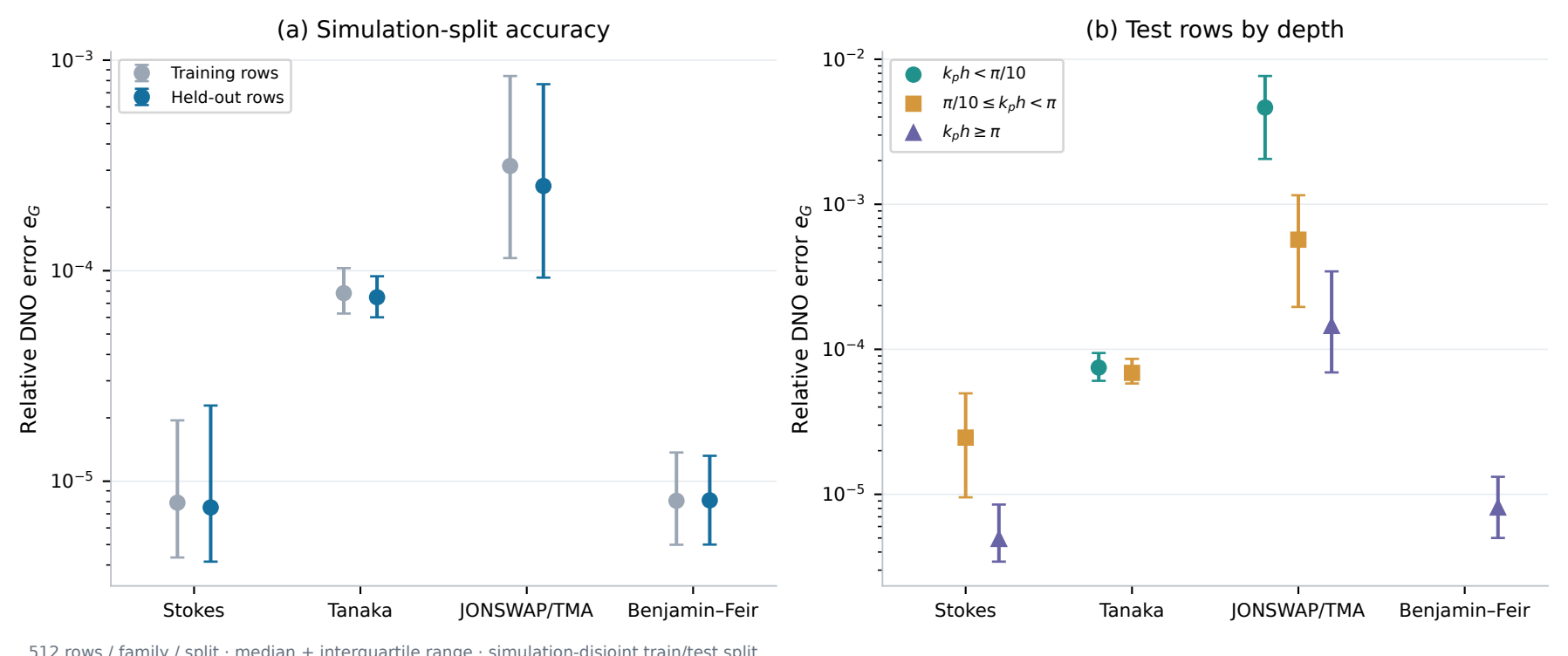
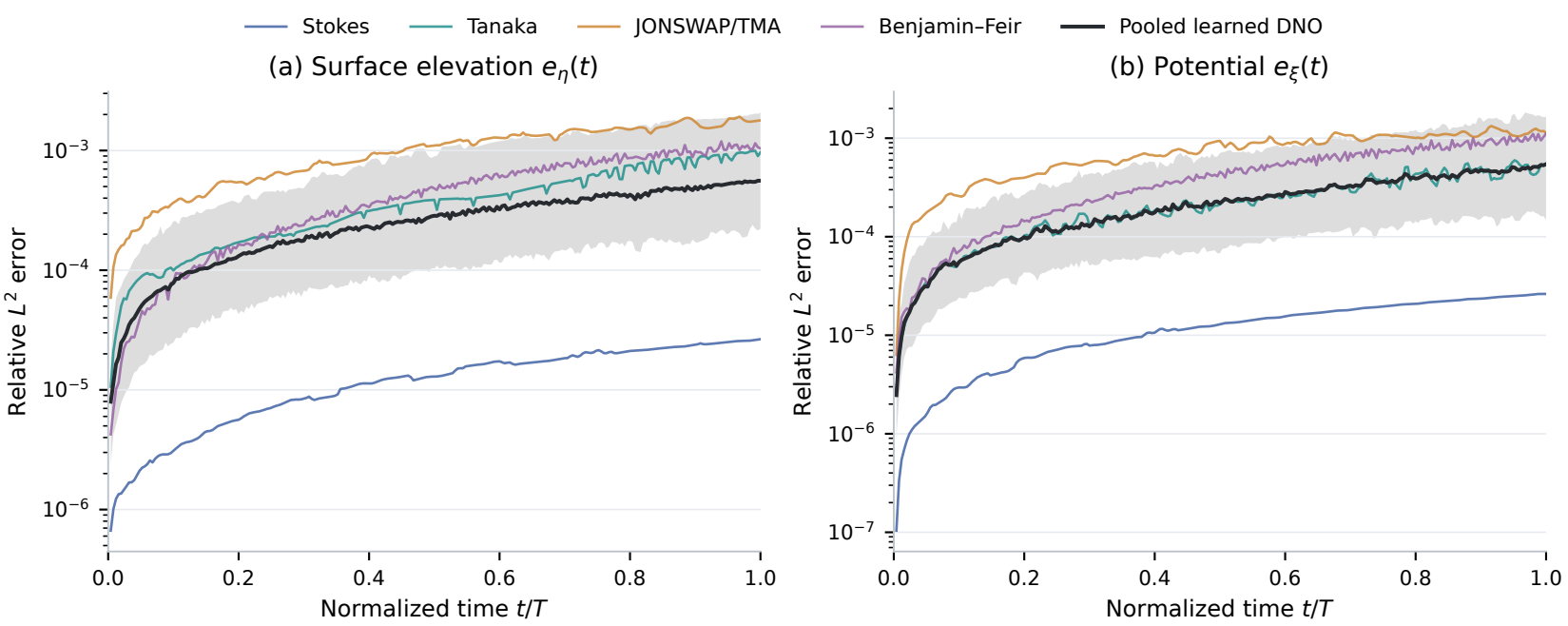
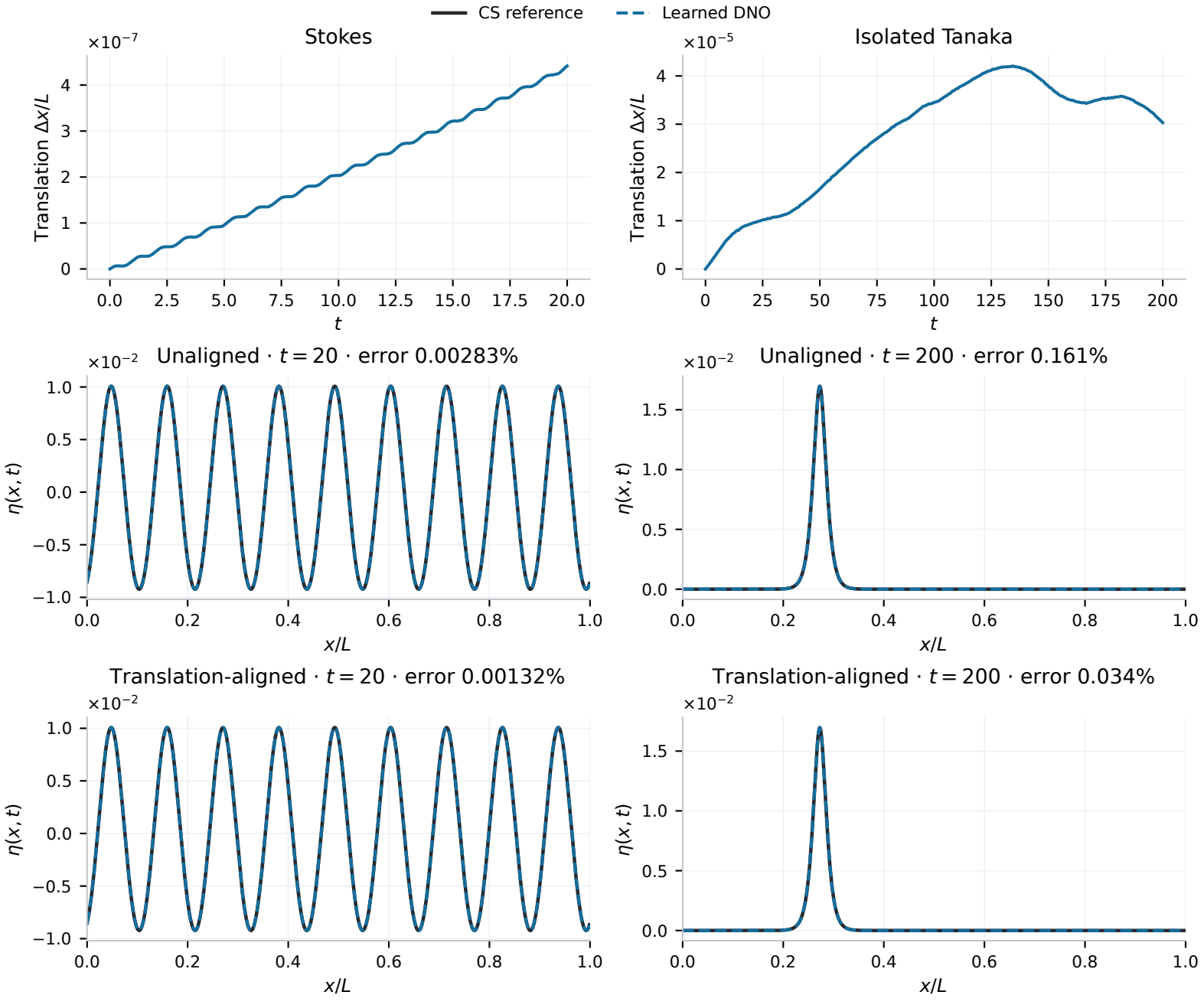


Training snapshots · cutoff at mode 128 · k_p from the surface spectrum · a = (max η − min η)/2

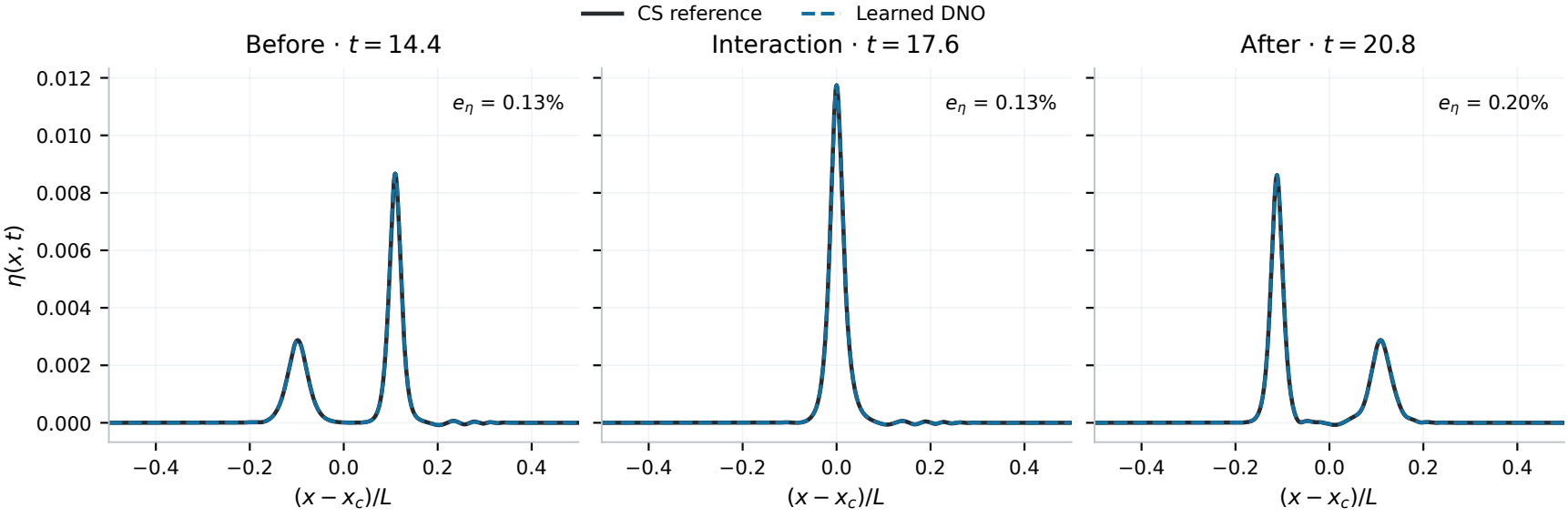

(a) Simulation-split accuracy
Training rows
Held-out rows
Relative DNO error e_G
10^{-3}
10^{-4}
10^{-5}
Stokes
Tanaka
JONSWAP/TMA
Benjamin–Feir
(b) Test rows by depth
$k_ph < \pi/10$
$\pi/10 \le k_ph < \pi$
$k_ph \ge \pi$
Relative DNO error e_G
10^{-2}
10^{-3}
10^{-4}
10^{-5}
Stokes
Tanaka
JONSWAP/TMA
Benjamin–Feir
512 rows / family / split · median + interquartile range · simulation-disjoint train/test split



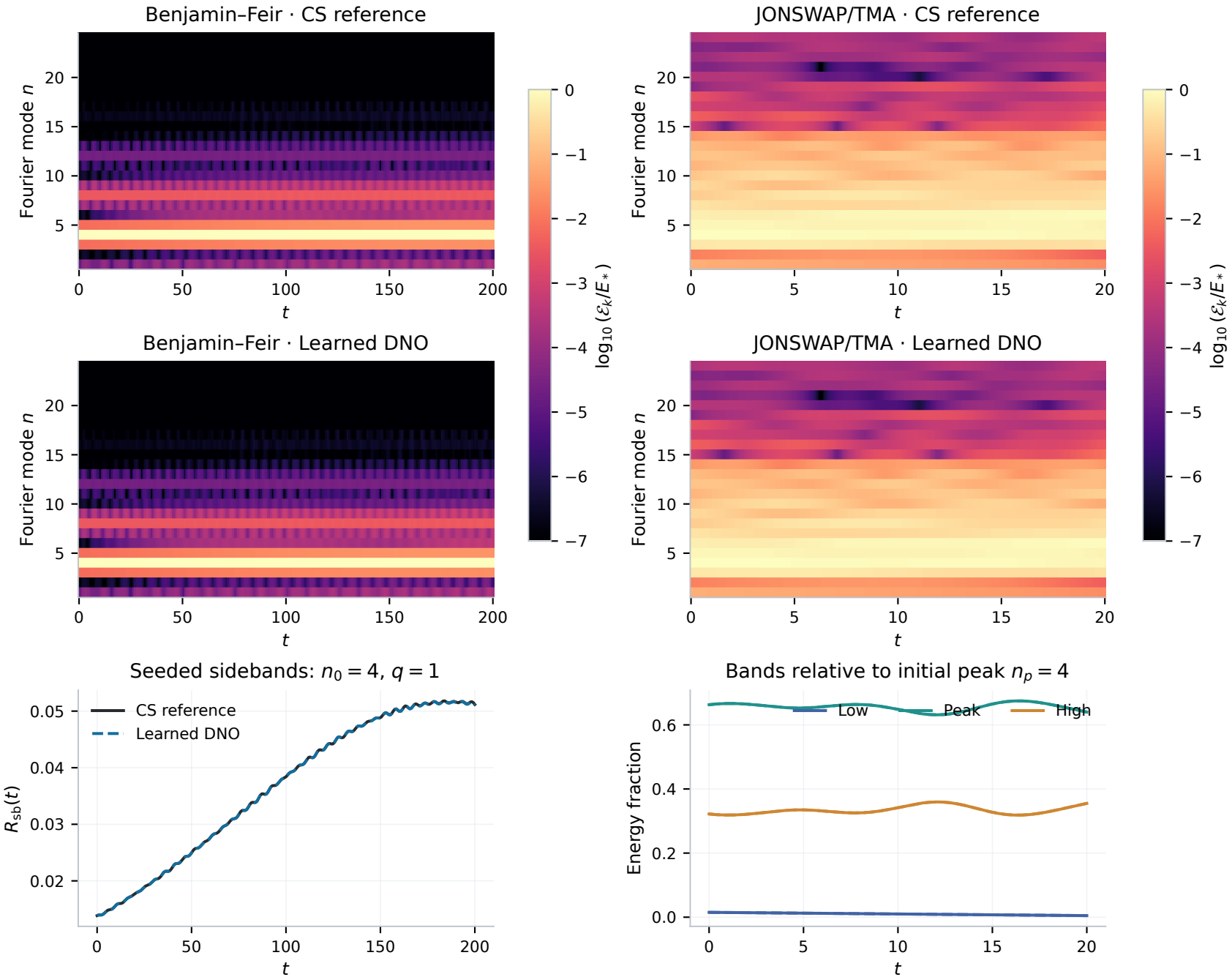
Learned DNO · 128 held-out simulations · no adaptive damping · N = 1024 · g = 1 · T = 20 or 200; normalized time



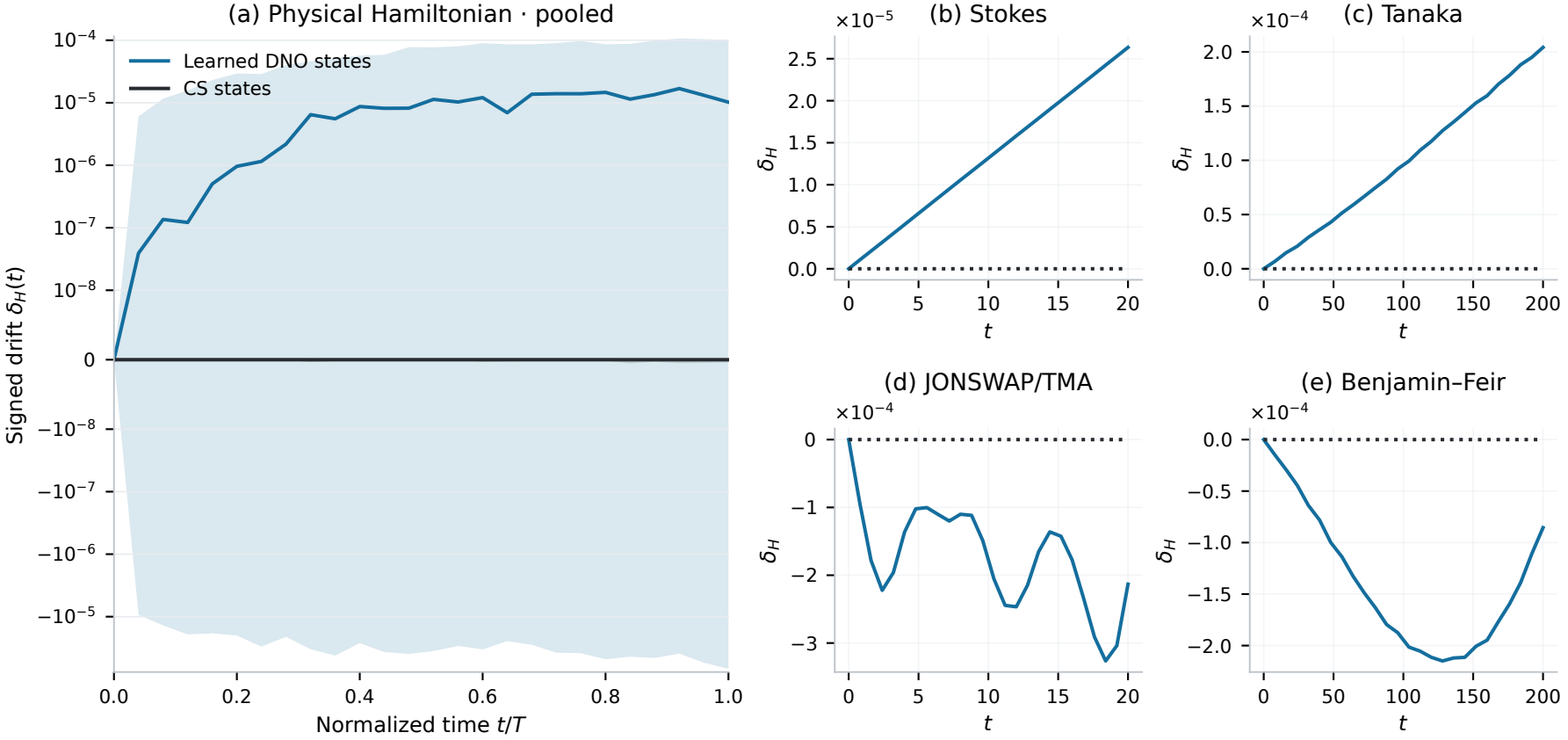
Top: continuous Fourier alignment · middle: raw prediction · bottom: aligned prediction · held-out test cases

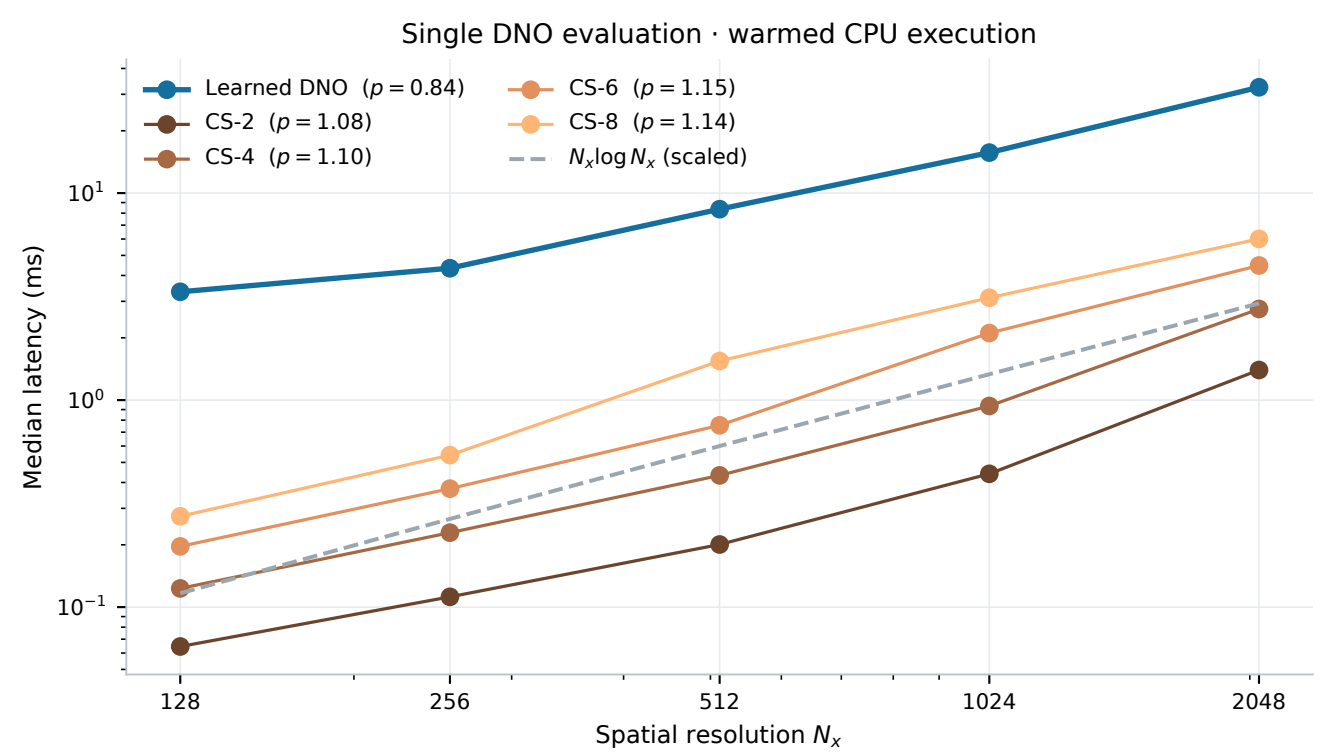


Two-wave Tanaka interaction · fixed recentering of both methods · event selected from reference crest amplification

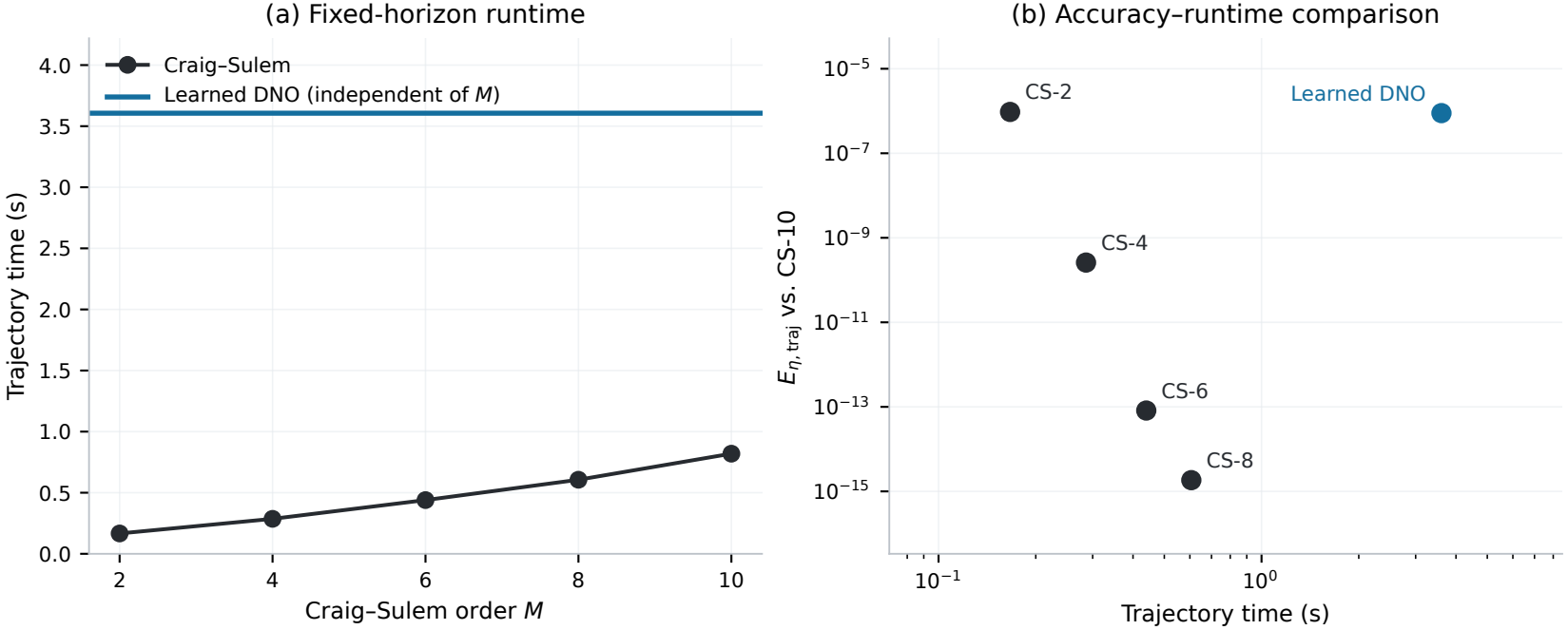


Matched color scales within each column · sea bands: solid CS / dashed learned DNO · g = 1


(a) Physical Hamiltonian · pooled
Learned DNO states
CS states
Signed drift $\delta_H(t)$
Normalized time t/T
(b) Stokes
(c) Tanaka
(d) JONSWAP/TMA
(e) Benjamin–Feir
δ_H
t
H evaluated with CS-6 at predicted states · 26 times × 128 cases · shaded bands = IQR



AMD EPYC 9124 CPU · 8 logical CPUs · FP32 net / FP64 physics · batch 1 · 7 repeats · JIT excluded
CS timings refreshed; saved neural measurements retained



CPU · same GL2 integration path · N = 256 · T = 1 · Δt = 0.01 · 3 repeats · compile time excluded
CS timings refreshed; saved neural measurements retained